Class Index No

Candidate Name: ______________________________

millennia institute

Promotional Examination 2008
Pre-university 2

BIOLOGY HIGHER 1
PAPER 2 **8875/2**

Thursday 11 September 2h

Additional materials:
Writing Paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.
Write your name, index number and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer **one** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Section A
Answer **all** the questions

1 **Figure 1** shows an electron micrograph of a human lymphocyte.

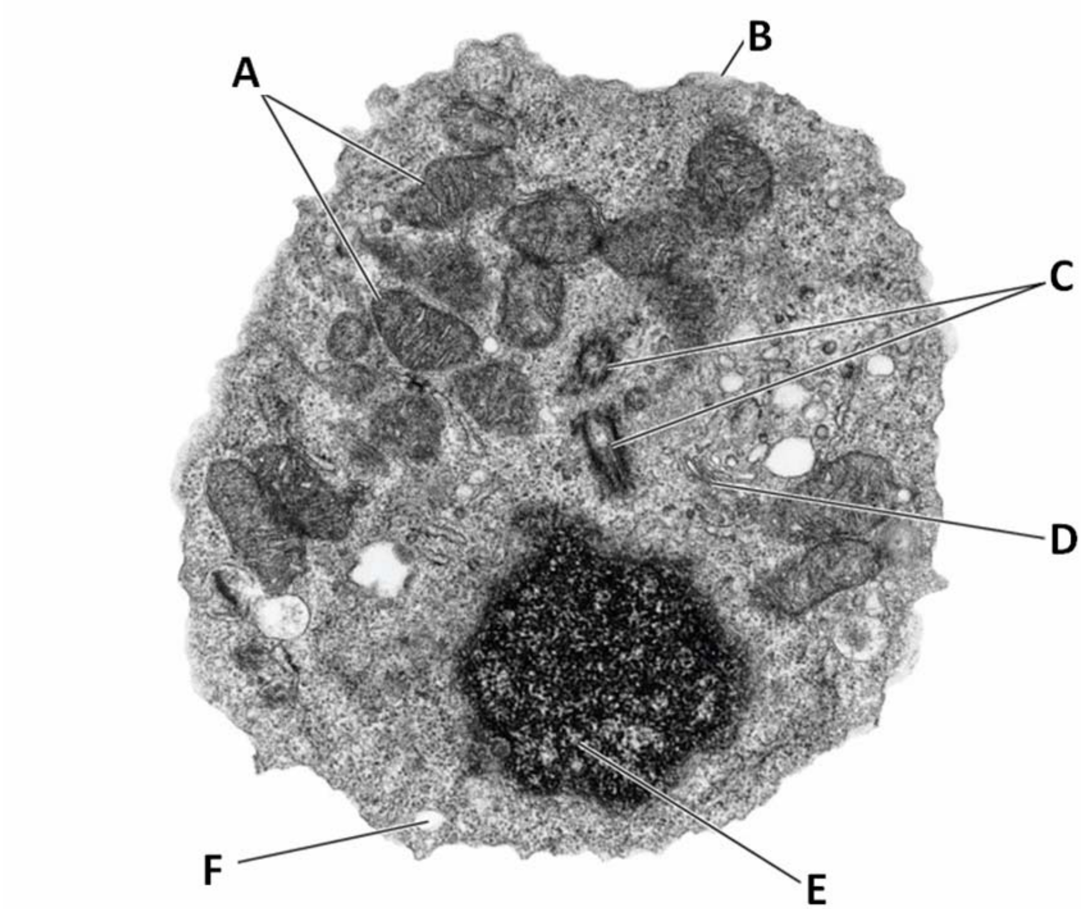

Figure 1

a. State what you will expect to find in E.

………………………………………………………………………………………………………

……………………………………………………………………………………………...…….. **[1]**

b. A cellular process takes place exclusively in structure A. Identify this process and state the raw materials of this process.

……………………………………………………………………………………………...............

……………………………………………………………………………………………………

……………………………………………………………………………………………...[2]

c. The function of D is to process and package proteins and lipids. Name the organelles that are responsible for the synthesis of these proteins and lipids.

……………………………………………………………………………………………………

……………………………………………………………………………………………………

…………………………………………………………………………………………………**[2]**

d. The function of structure B is to act as a barrier between the cytoplasm and the extracellular environment. The main component of B is an amphipathic molecule. In the space below, draw a labelled diagram to illustrate the behaviour of the amphipathic molecule when they are put in water.

[2]

e. Structure B is not permeable to charged particles and polar molecules. State with examples how these particles are transported across the membrane.

....

....

....

....

....**[3]**

f. State a cellular activity which C is involved in.

....

....**[1]**

[Total: 11]

2 **Table 1** shows the relative amounts of the bases Adenine, Thymine, Guanine and Cytosine in DNA from different organisms.

Source	*adenine*	*thymine*	*guanine*	*cytosine*
Bacterium	23.8	23.1	26.8	26.3
Maize	26.8	27.2	22.8	23.2
Fruit fly	30.7	29.5	19.6	20.2
Chicken	28.0	28.4	22.0	21.6
Human	29.3	30.0	20.7	20.0

Table 1

a. Explain the importance of the ratios of A to T and G to C to the structure of DNA.

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……………………………………………………………………………………………**[3]**

b. Describe the role of the following proteins in replication.

i. Helicase

……………………………………………………………………………………….....

....**[1]**

ii. Single-strand binding protein

…………………………………………………………………………………………

…………………………………………………………………………………………

…………………………………………………………………………………………**[2]**

c. DNA polymerase synthesises DNA in replication while RNA polymerase synthesises RNA during transcription. State another difference between DNA polymerase and RNA polymerase.

..

...[1]

d. State one difference between a pre-mRNA and a matured mRNA.

..

...[1]

e. Name the parts of a ribosome that is responsible for its role in translation.

..

..

...[2]

[Total: 10]

3 In the garden pea, *Pisum sativum*, two pairs of alleles determine the seed characters green and yellow and round and wrinkled. **Table 2** shows the results which were obtained from four separate crosses.

cross	**parent**	**Progeny**			
		Yellow round	**Yellow wrinkled**	**Green round**	**Green wrinkled**
1	Yellow round x green wrinkled	138	143	137	142
2	Yellow round x yellow round	176	0	60	0
3	Yellow round x yellow wrinkled	223	247	75	86
4	Green round x yellow wrinkled	371	0	0	0

Table 2

a. State which alleles are dominant and explain your answer.

....

....

....**[2]**

b. Using one of the crosses explain whether the loci for the genes for the seed colour and the seed shape are located on the same chromosome or different chromosome.

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....**[4]**

c. Explain the following observation:

i. Color blindness is a genetic disorder that is more predominant in males than in females.

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....**[2]**

ii. A couple, one with blood group A and the other blood group B has children with blood groups A, B and O.

....

....

....**[2]**

[Total: 10]

4 Cells generate ATP by adding a phosphate group (P_i) to ADP. During the complete oxidation of glucose, cells have two ways of doing this:

- Substrate level phosphorylation
- Oxidative phosphorylation

Figures 2 and 3 are diagrams that show the main details of these two processes (not drawn to the same scale).

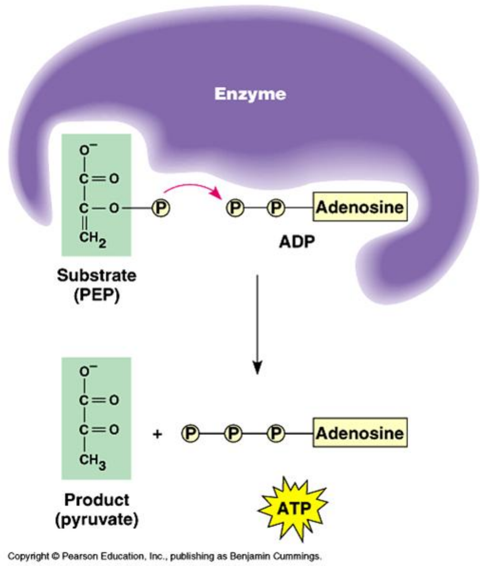

Figure 2

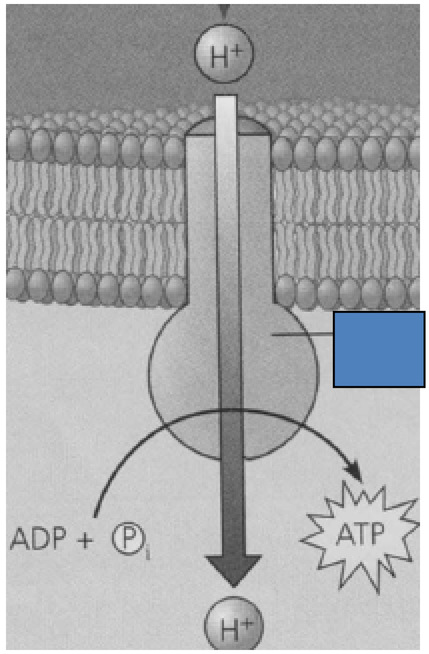

Figure 3

a. Describe the importance of ATP in cells, giving two examples of processes in which it is used.

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....

....

....**[3]**

b. State precisely where these two processes occur in a cell.
Substrate level phosphorylation
Oxidative phosphorylation

....

....

....**[2]**

c. Compare the relative amounts of ATP produced by the two processes when a molecule of glucose is completely oxidised.

....

....**[1]**

d. Only substrate level phosphorylation is possible in the absence of oxygen. Explain why oxidative phosphorylation is not possible in the absence of oxygen.

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....**[3]**

Section B

Answer only **one** question

Marks will be awarded for a detailed, broad and balanced account.

5(a) Describe the four levels of protein structure. State one feature of the enzyme HIV-1 protease to illustrate each level of protein structure. **[10]**

(b) What are the benefits of the Human Genome Project? **[10]**

[Total: 20]

6(a) Distinguish between competitive and non-competitive inhibition of enzymes. **[6]**

(b) Describe the role of light in photosynthesis. **[8]**

(c) Explain the similarities and differences between homologous chromosomes. **[6]**

[Total: 20]

END OF PAPER

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